



SEMESTER 2 EXAMINATIONS 2021/2022

MODULE: CA4024 - Building Complex Computational Models

PROGRAMME(S):
DS BSc in Data Science

YEAR OF STUDY: 4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[4 Marks]**

In forecasting, the predicted value of the dependent variable y at time t denoted $\hat{y}_t$, can be made using the regression: $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t + \hat{\beta}_2 x_t^2 + \hat{\beta}_3 x_t^3 + \dots$. The values of the unknown coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$ can either be estimated *ex post* or *ex ante*. Explain carefully what is the difference between *ex post* and *ex ante* forecasting in time series analysis.

Q 1(b)**[12 Marks]**

The time series data in Table Q1 shows Quarterly Sales Data (in Millions of Tons) from 2004-07. Generate a graph of the data explaining what kind of model should be used to decompose the data and provide reasons for your choice. Calculate the Trend (using 2X4pt Moving Averages), Seasonal Index and Seasonally-Adjusted data. Graph the Trend and Seasonally-Adjusted data (but **not** the Seasonal Index data).

Year	Quarter	Sales
2004	1	11
2004	2	12.5
2004	3	22
2004	4	15
2005	1	12.5
2005	2	13
2005	3	23
2005	4	15.5
2006	1	14
2006	2	15.5
2006	3	24.5
2006	4	17.5
2007	1	15
2007	2	17
2007	3	27
2007	4	21

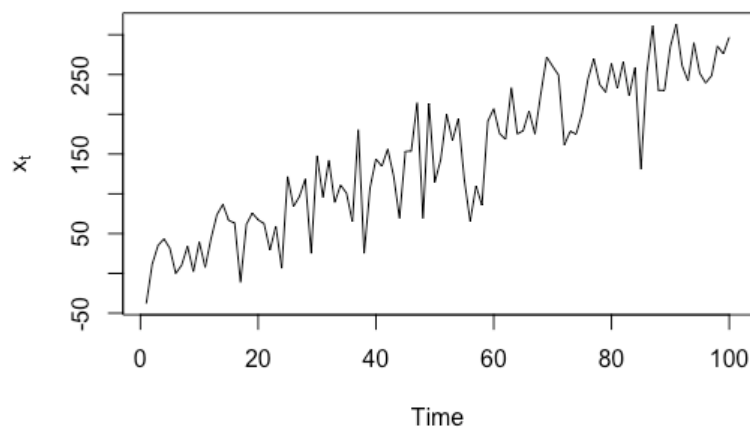
Table Q1**Q 1(c)****[9 Marks]**

Using the results in Q1(b), generate the forecast data for 2008. Comment on whether this is an *ex post* or *ex ante* forecast.

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]****Q 2(a)****[16 Marks]**

Explain carefully what is meant by rendering a time series (a) trend stationary and (b) difference stationary. For the graph in Fig Q2(a), explain how one of these techniques may be used to render it stationary. Draw a rough sketch of the result of applying such a technique. Draw rough sketches of the Auto Correlation Function (ACF) of the function in Fig Q2(a) and the detrended graph.

**Fig Q2(a)****Q 2(b)****[9 Marks]**

For the series shown in Fig Q2(b), identify, giving your reason, which series are stationary.

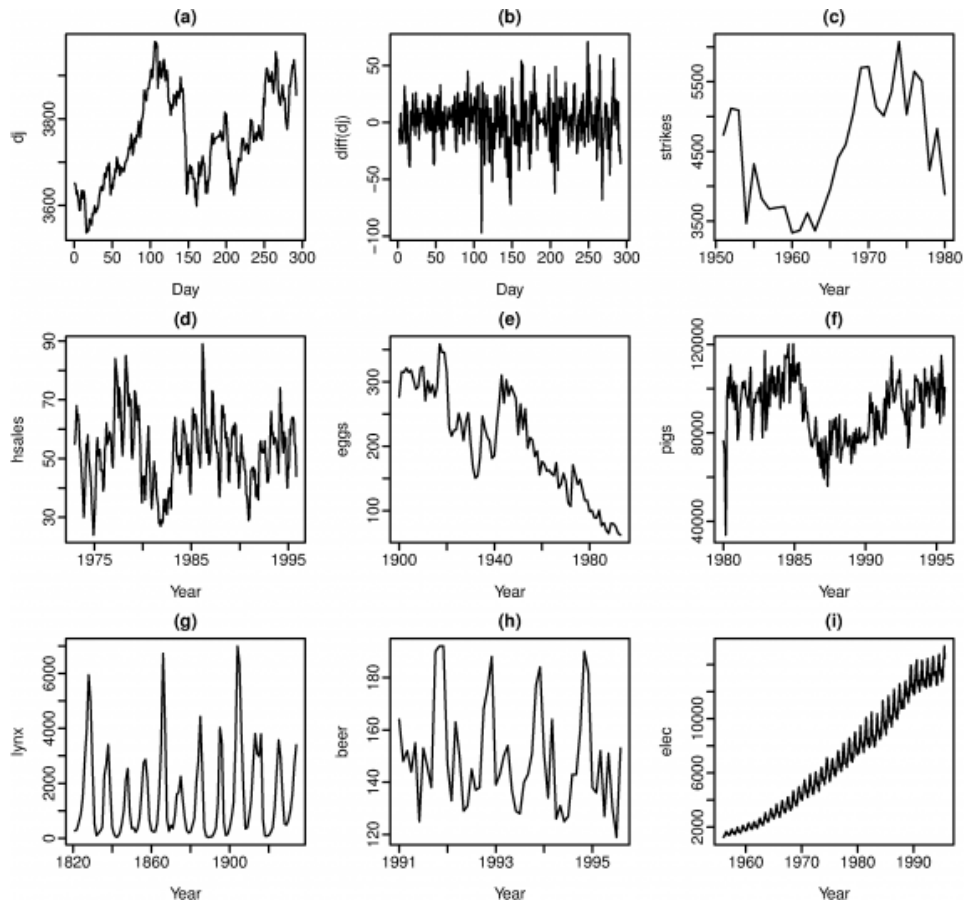


Figure Q2(b)

[End of Question2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[8 Marks]

Give your own explanations and examples of:

- i) Emergent behaviour
- ii) Self-organisation

What is the difference between the two?

Q 3(b)

[9 Marks]

What is considered a good model in complex systems? Give your own example of a good model design explained through typical modelling steps.

Q 3(c)

[8 Marks]

The following figures show the bifurcation diagram of the logistic map, r along the x-axis. Please give clear explanations for what is happening at critical points for r . For values above 3.5 use a bottom figure i.e. **Figure Q 3(c) (down)** with zoomed in graph.

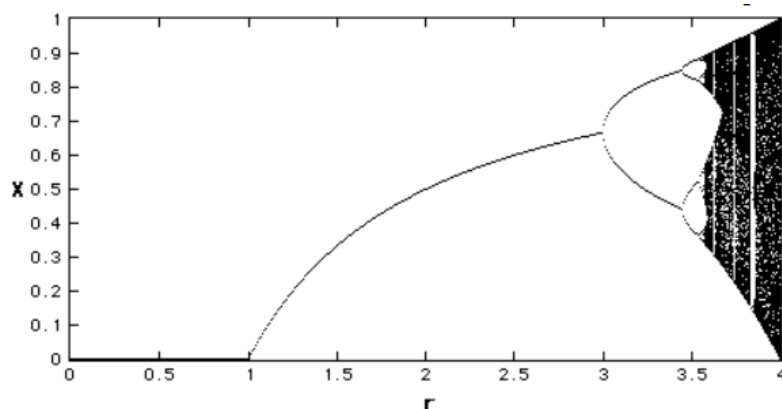


Figure Q 3(c) (up)

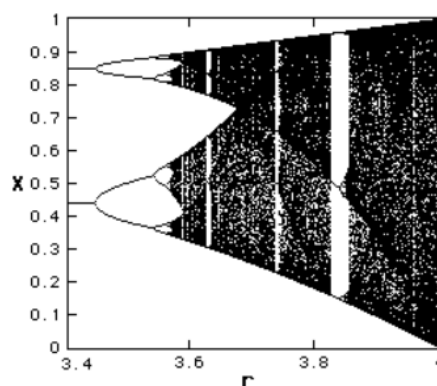


Figure Q 3(c) (down)

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a) [6 Marks]**

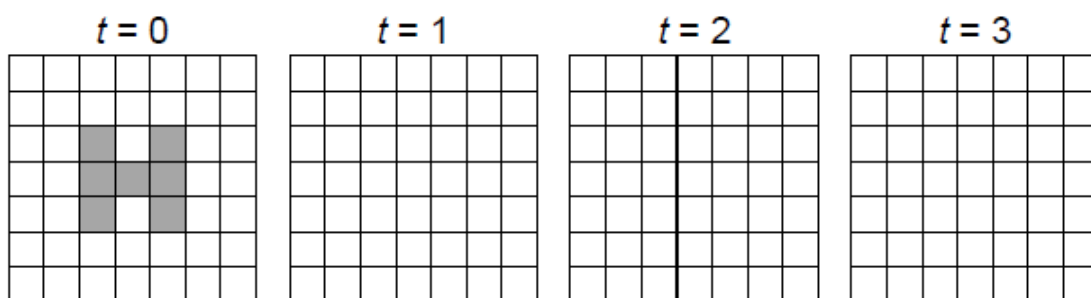
- (i) What are the typical properties of agents in Agent-Based Modelling (ABM)?
- (ii) Using an example of birds flocking, draw main rule sets showing how they can be used in ABM. (You can use similar logic as e.g. in boids example).

Q 4(b) [5 Marks]

Using a method of your choice (e.g. Louvain method, Girvan-Newman method, random walk method or other) explain concisely how it can be used to find optimal community structure that maximises the modularity.

Q 4(c) [6 Marks]

Figure below is an example of a 2D Cellular Automata (CA) model with von Neumann neighbourhood and with *no boundary* (infinite space). White means 0, gray means 1. Each cell switches to $\text{round}(S/5)$ in every time step, where S is the local sum of states within its neighbourhood. Complete the time evolution of this CA.

**Q 4(d) [8 Marks]**

Using a simple 2D forest fire example model (or another such application from your knowledge), draw a schematic illustration of the model's state-transition function. Clearly define all possible states and write what typical properties of CA can be seen in this example.

What would happen if the space of the forest fire propagation were 1-D or 3-D?

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 25]****Q 5(a)****[4 Marks]**

What is a Markov Process? What are the main assumptions of a Markov Process?

Q 5(b)**[8 Marks]**

Recent storms like Dudley and Eunice can be categorized in several categories based on the strength of wind: *Category 1* (33-42m/s), *Category 2* (43-49m/s), *Category 3* (50-58m/s), *Category 4* (58-70m/s).

- Assume the following transition probabilities from the *Category 1* state: *Category 2* (0.75), *Category 3* (0.2), *Category 4* (0);
 - Assume the following transition probabilities from the *Category 2* state: *Category 1* (0.25), *Category 2* (0.3), *Category 3* (0.45);
 - Assume the following transition probabilities from the *Category 3* state: *Category 2* (0.4), *Category 3* (0.3), *Category 4* (0.2);
 - Assume the following transition probabilities from the *Category 4* state: *Category 2* (0.15), *Category 3* (0.3), *Category 4* (0.55).
- (i) Draw a Markov transition diagram and associated transition matrix for this system.
- (ii) From your diagram in Q5b(i) find the probability that *Category 2* storm will transition to *Category 1* after 1, 2 time steps.

Q 5(c)**[5 Marks]**

Explain concisely what is meant by a Markov Chain Monte Carlo (MCMC) and when it should be used.

Q 5(d)**[8 Marks]**

Explain MCMC general algorithm and workflow in terms of prior and posterior belief. Specifically demonstrate what is the role of a Sampler in MCMC algorithm (choose a Sampler of your choice highlighting its advantages).

[End of Question 5]**[END OF EXAM]**